

Accessible VR Museum Experience Tailored to User Needs

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ABSTRACT

This study aims to investigate and discuss methods for increasing accessibility in virtual reality (VR) applications to provide an inclusive immersive experience. Indeed, accessible VR promotes the setting-up of scenarios that can be adapted in all their aspects according to users' needs, in order to overcome issues deriving from sensory and motor impairments, as well as from cultural differences. In particular, in this contribution, these methods are applied to design and develop a virtual museum experience, where both the environment and interactions are adapted to the visitors' physical and personal information. This customization is aimed at enhancing comfort, ease of use, and intuitiveness, ensuring that the experience meets each user's specific needs and preferences.

Index Terms: Virtual Reality, accessibility, inclusion, VR museum.

1 INTRODUCTION

Virtual Reality (VR) stands out for its ability to immerse users in entirely virtual environments and explore them. To ensure these experiences are accessible and inclusive, VR environments must address the needs of diverse users groups, including individuals with disabilities, varying ages, and different cultural backgrounds. By applying universal design principles long used in physical museums, developers can create virtual environments that accommodate a wide range of abilities. Examples include using clear and distinguishable fonts and colors, leveraging audio to complement text, and positioning interactive elements for ease of access. Guidelines from W3C consortium [8] and the XR association [9] emphasize the importance of adaptable VR environments to meet these diverse needs. Significant work has been done to address inclusivity in physical museums, but equivalent progress in VR applications remains necessary. Research efforts include non-visual VR solutions that use audio and haptic feedback for visually impaired users [10], as well as immersive adaptations for wheelchair users who face physical access challenges [5]. Additional work has been dedicated to creating VR experiences for children with disabilities, particularly those on the autism spectrum [2]. However, tailored solutions that address the diverse requirements of heterogeneous groups of people remain an area for growth. Building on these insights and recent research [3, 7, 9], our work presents a VR application that dynamically adapts to the specific needs of individual users. This research is conducted as part of the PNRR Project RAISE Spoke 1, which focuses on developing inclusive technologies to improve the quality of life for all citizens. Within this framework, based on experimentation performed with users of different cultures and ages, we propose an immersive virtual museum experience that adjusts its settings to enable seamless interaction with virtual objects, providing an accessible and engaging environment for all users.

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Embracing the “design-for-one” philosophy, this approach aims to deliver deeply automatically personalized experiences that address the unique preferences and requirements of each individual. Key features include adaptable locomotion options, optimized content presentation, and flexible interaction methods, establishing a solid foundation for future innovations in inclusive VR design.

2 METHODOLOGY

The system aims to adhere to accessibility principles such as equitable use, flexibility in use, perceptible information, and size and space for approach and use [4]. It addresses various motor and sensory impairments by providing a personalized scene adjustment tailored to individual user needs and two core accessibility features.

In the following, the design and setup of a VR scene aimed at creating an accessible and inclusive environment for exploring a virtual museum exhibition is outlined. The scene is a 3D reconstruction of a real museum area where pedestals are arranged to display artifacts, which are digital replicas of archaeological remains. To leverage VR's potential to enhance experiences and provide augmented information, each pedestal is equipped with two interactive buttons: one for displaying text and the other for playing audio.

2.1 Scene Personalization

Personalization is a key feature of the system, allowing the VR scene to adapt to user-specific parameters, such as height, physical abilities, or cultural backgrounds. One of the ultimate goals, as part of the PNRR Project RAISE Spoke 1, is to develop a digital citizen model to enable automated personalization across various applications, including the one here presented, enhancing inclusivity and well-being. While future implementations aim for fully automated settings, the current system also collects essential information through intuitive prompts, as described in the following.

Language Settings. Language barriers are addressed by offering multiple language options, ensuring an inclusive experience for all users. The language selection process uses visual aids and simple concepts. First a panel displays a world map for continent selection, then flags of countries within the selected continent are shown, labeled with their official languages (e. g., to avoid confusion between flags, even in the case of color blindness and in case of multiple languages countries). This feature makes information perceptible and ensures the system's flexibility for diverse cultural and linguistic groups.

Color Customization. Color blindness, a common accessibility challenge in reality, can be mitigated in VR through customizable color schemes. For instance, VR buttons typically change colors as feedback to indicate their activation/deactivation, using red and green as default colors, but this is a problematic combination for many. As a result, users can select pairs of complementary colors or pairs of textures for interactive elements. These have been chosen based on an analysis of studies on color blindness and the inclusion of textures allows a higher likelihood that everyone can distinguish at least one pair. This feature ensures that buttons and other components are visible and operable regardless of sensory impairments.

Object Positioning. The physical arrangement of objects can create barriers for users with mobility impairments or height. These barriers are here overcome by customizing objects positioning to

suit the user's physical dimensions. That is, pedestal heights are adjusted automatically based on the user's height. Taking into account the environmental mapping characteristics of Meta Quest and the standard human body proportions, the top of each pedestal is aligned with the user's elbow height, ensuring easy access to artifacts and buttons (Fig. 1). This dynamic setup enhances accessibility, equity in use, and the comfort of the experience for all users, ensuring an appropriate space configuration.

2.2 Accessibility Features

The accessibility features include two locomotion techniques and customizable content fruition. These are detailed in the following.

Locomotion Techniques. Two locomotion techniques [1] are provided to accommodate diverse physical abilities, including users using wheelchairs, having mobility impairments, or elderly people. Both techniques are inspired by real-world movements to create a natural experience in the metaverse and to foster the sense of presence [6]. The elimination of intermediary devices has to be underlined, that enhances realism and simplify the interactions. The first is a *natural walking method*, which leverages the headset's room-scale VR tracking to allow users to physically walk within the virtual environment, mirroring real-world movements. This immersive method is ideal for users without significant mobility limitations, and it can also be suitable for those with mobility issues in the upper limbs. The second is a *semi-natural pointing method*, which enables users to navigate the virtual environment by pointing in the desired direction while remaining stationary. This option is particularly suitable for seated users, those with mobility impairments, or balance problems. It ensures that users can fully explore the environment without physical effort and can be seen as a means for escaping reality and enjoy the experience. Both techniques can be used interchangeably, allowing users to switch between them based on their preferences, needs, and how they perceive the experience, such as in cases of cybersickness. This dual-method approach guarantees accessibility, providing a comfortable and inclusive experience for everyone navigating the virtual museum.

Content Fruition: Text and Audio. The information associated to the various artifacts enhances the user experience and is designed to be accessible to everyone. Each pedestal features two buttons: one to display a text panel and another to play an audio description, both in the previously selected language. Users with audio impairments can select textual information displayed on a highly visible panel at eye level, ensuring clear readability. For users with reading disabilities, such as dyslexia or visual impairments, audio descriptions can be played, providing an auditory alternative. By offering both text and audio formats, the system ensures content is perceivable and accessible to users with varying needs. A key feature is the inclusion of all languages, made possible by a developed asynchronous translation and text-to-speech conversion tool. This dual-mode content fruition, combined with the ability to interact with the artifacts, by grasping, rotating, and zooming them, further enriches the overall experience, making it more engaging and tailored to individual preferences.

3 CONCLUSION

Despite of their inclusiveness potentiality, most VR applications are designed either for standard users or specific categories of physically impaired individuals, limiting their broader inclusivity potential. To address these limitations, the paper presents the development of an immersive, accessible, and user-centered virtual museum exhibition that allows users to freely explore the environment and access the information content in most appropriate way. Methods have been implemented to tailor the rendering of information and the interactions to accommodate moderate physical impairments as well as differences in age and cultural backgrounds. These methods can support virtual museums and cultural heritage exhibi-

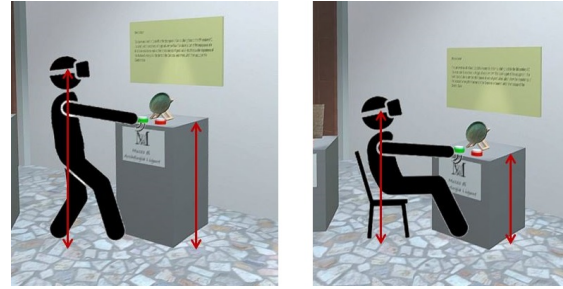


Figure 1: Examples of objects positioning based on user height: user standing and user seated.

tions in their significant educational and inclusive value, breaking down economic and physical barriers to access physical settings. The initial testing of the system provided positive feedback. In the future, we plan to carry out further testing with a more diversified and impaired person sample to fully assess accessibility, while also striving to promote the collaborative aspect of end-users in co-design processes.

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